

@stacks/blockchain-api-client

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npm v8.11.5

A JS Client for the Stacks Blockchain API

Features

This package provides the ability to:

- Execute REST API requests against the Stacks Blockchain API
- Subscribe to WebSockets or Socket.io for real-time Stacks updates (see [Available Updates](#))
- Full type safety for WebSocket and API requests and responses

Documentation

The documentation for the client library is published as [github pages](#).

Installation

You can install this package using NPM:

```
npm install --save @stacks/blockchain-api-client
```

Usage

Here is example code that subscribes to updates for a specific Stacks address:

Websockets

```
import { connectWebSocketClient } from '@stacks/blockchain-api-client';

// for testnet, replace with wss://api.testnet.hiro.so/
const client = await
connectWebSocketClient('wss://api.mainnet.hiro.so/');

const sub = await
client.subscribeAddressTransactions('ST3GQB6WGCWKDNFNPSQRV8DY93JN
06XPZ2ZE9EVMA', event =>
  console.log(event));
```

```
await sub.unsubscribe();
```

Socket.io

```
import { io } from "socket.io-client";
import * as stacks from '@stacks/blockchain-api-client';

// for testnet, replace with https://api.testnet.hiro.so/
const socketUrl = "https://api.mainnet.hiro.so/";

const socket = io(socketUrl, {
  transports: [ "websocket" ]
});
const sc = new stacks.StacksApiSocketClient(socket);

sc.subscribeAddressTransactions('ST3GQB6WGCWKDNFNPSQRV8DY93JN06XP
Z2ZE9EVMA');
```

Available Updates

Block Updates

Sent every time a new Stacks block is mined.

Example message:

```
{
  "canonical": true,
  "height": 3275,
  "hash":
    "0xe77ba8cf6bb7c0e4f64adc83356289ed467d31a22354907b4bb81459005843
    0f",
  "parent_block_hash":
    "0x75ab21ef25cbff2caa14c27d830ed7886a4d1522e1b6f9e5dc3b59ccf73ed4
    9f",
  "burn_block_time": 1594233639,
  "burn_block_time_iso": "2020-08-27T16:41:26.000Z",
  "burn_block_hash":
    "0xb154c008df2101023a6d0d54986b3964cee58119eed14f5bed98e15678e18f
    e2",
  "burn_block_height": 654439,
  "miner_txid":
    "0xd7d56070277ccd87b42acf0c91f915dd181f9db4cf878a4e95518bc397c240
    cc",
  "parent_microblock_hash":
    "0x500c1bb1d7bcbccfca0c0fca0f8c60c26040e103d14687fca05fba4da1c71d40
```

```

"parent_microblock_sequence": 2,
"txs": [

"0x4262db117659d1ca9406970c8f44fffd3d8f11f8e18c591d2e3960f40701077
54",

"0x383632cd3b5464dfffb684082750fcfadd1f52625bbb9f884ed8f45d2b1f05
47",

"0xc99fe597e44b8bd15a50eec660c6e679a7144a5a8553d214b9d5f1406d278c
22"
],
"microblocks_accepted": [

"0xce0b1a4099d3fc7d5885cc7a3baa952b6d999f9709d0683b98b84359720823
1c",

"0x4c0529b6448a5885991c5021bd869cc97f1692c128a98b382729dc962203c3
26",

"0x64968846291dfea1015228a9d4bbd60aac81378cd6774b810b08e59e6b0e74
94"
],
"microblocks_streamed": [

"0xb5650ef855f7d90fc146942e85cf9fac3a8c47ec408aca02f3cf9ed7c82f6c
c6",

"0xeeb9aa5741d84aa0bc5de4f2fbdeae57ae29694479475d45a67ae7bd7e2c98
f3",

"0x4f4c368d5f06fdf6065c5bafd9cb37391fddc9c279cfc57be35e4bf8ee932c
bd",

"0xde2fc8d99872c827f144c752c002d29f9315dfc09472a09572ac7447ae623d
ea"
],
"execution_cost_read_count": 2477,
"execution_cost_read_length": 1659409,
"execution_cost_runtime": 2520952000,
"execution_cost_write_count": 608,
"execution_cost_write_length": 80170
}

```

Subscribe via WebSockets:

```
client.subscribeBlocks(event => {});
```

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Subscribe via Socket.io:

```
sc.subscribeBlocks();
```

Microblock Updates

Sent every time a new Stacks microblock is streamed.

Example message:

```
{
  "canonical": true,
  "microblock_canonical": true,
  "microblock_hash":
    "0xa31ee2244ceee0d042c0b129a91df2433c4ffd3b94e7e4e5dfa3a15927684a
    6f",
  "microblock_sequence": 0,
  "microblock_parent_hash":
    "0x5d053c206a7bcc5dcfe8bb8a61d8699fc068179f388eb2aec62786b7318c36
    c4",
  "block_height": 38224,
  "parent_block_height": 38223,
  "parent_block_hash":
    "0x5d053c206a7bcc5dcfe8bb8a61d8699fc068179f388eb2aec62786b7318c36
    c4",
  "block_hash": "",
  "txs": [

    "0xe4b46358b7864c9db31e15e7db4f74042a2e1748db920b93480ed56463ac1c
    48",
    "0x1de11ca776fc4a713465bc5974790cd3cdab7d8ad89fa474c10a4160cf89ef
    dc",
    "0x16b3a99d6d100562964f6f0f0ae02d47121386495c4fd903e4e9f72548ae0b
    35",
    "0xc99f802dffffee9190e4b8ee5c128295f4eec51974d7cabdec0ba49c35b17bc
    a5",
    "0xeffa0b1d1e5b96dc1de1ff49b390752f35cbe49eddf5a6c7ff1ee80ea2b738
    86",
    "0x93f927bdf15056f65ff3b0c6041287c9cfd070e4bddadedeb1aa1627d83702
    2a",
    "0xdbd62060daf483a7fdac2d76c3a88091690ff1f26827e2627c0894e5f73dcf
```

```
"0x0d89a6edb51f96eec1dfbbaf1abc66861c96c75a6962545c0783d78773563a
4b"
],
"parent_burn_block_height": 710158,
"parent_burn_block_hash":
"0x00000000000000000000000007b6fa2dcd91e0c69d488f9742d7e5261286aefce29e
e0",
"parent_burn_block_time": 1637167098,
"parent_burn_block_time_iso": "2021-11-17T16:38:18.000Z"
}
```

Subscribe via WebSockets:

```
client.subscribeMicroblocks(event => {});
```

Subscribe via Socket.io:

```
sc.subscribeMicroblocks();
```

Mempool Updates

Sent every time a new transaction is submitted to the mempool. Transactions of different types and structures may be received.

Example message:

```
{
  "tx_id":
  "0x698096b646d17297836542beecfe8d6a30454b2ce5ce4e9b0e5d243b05dce9
98",
  "nonce": 66,
  "fee_rate": "554000",
  "sender_address": "SPBDCE0KCWY56SX6XE7MB6DVVETABYM2WRDK6PCB",
  "sponsored": false,
  "post_condition_mode": "allow",
  "post_conditions": [],
  "anchor_mode": "any",
  "tx_status": "pending",
  "receipt_time": 1637172201,
  "receipt_time_iso": "2021-11-17T18:03:21.000Z",
  "tx_type": "contract_call",
  "contract_call": {
    "contract_id":
    "SP2C2YFP12AJZB4MABJBAJ55XECVS7E4PMMZ89YZR.arkadiko-swap-v2-1",
    "function_name": "swap-x-for-y",
```

```

    "function_args": [
      {
        "hex":
"0x0616982f3ec112a5f5928a5c96a914bd733793b896a50e61726b6164696b6f
2d746f6b656e",
        "repr":
"SP2C2YFP12AJZB4MABJBAJ55XECVS7E4PMMZ89YZR.arkadiko-token",
        "name": "token-x-trait",
        "type": "trait_reference"
      },
      {
        "hex":
"0x0616982f3ec112a5f5928a5c96a914bd733793b896a50a757364612d746f6b
656e",
        "repr": "SP2C2YFP12AJZB4MABJBAJ55XECVS7E4PMMZ89YZR.usda-
token",
        "name": "token-y-trait",
        "type": "trait_reference"
      },
      {
        "hex": "0x01000000000000000000000000000003b023380",
        "repr": "u990000000",
        "name": "dx",
        "type": "uint"
      },
      {
        "hex": "0x01000000000000000000000000000062ea4a9f",
        "repr": "u1659521695",
        "name": "min-dy",
        "type": "uint"
      }
    ]
  }
}

```

Subscribe via WebSockets:

```
client.subscribeMempool(event => {});
```

Subscribe via Socket.io:

```
sc.subscribeMempool();
```

Transaction Updates

Sent every time a single transaction (subscribed by transaction ID) is updated.

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Example message if subscribed to updates for a transaction with ID

0xd78988664aaa9a1b751cd58c55b253914f790e95ca6f3d402a866559e1cbe0b3
after it was submitted to the mempool:

```
{
  "tx_id":
  "0xd78988664aaa9a1b751cd58c55b253914f790e95ca6f3d402a866559e1cbe0b3",
  "nonce": 18,
  "fee_rate": "74400",
  "sender_address": "SP36ADRBVM8J00ZWR5QXC8V65WTJNCD1BF4EJ93ZZ",
  "sponsored": false,
  "post_condition_mode": "deny",
  "post_conditions": [
    {
      "type": "stx",
      "condition_code": "sent_less_than_or_equal_to",
      "amount": "50000000",
      "principal": {
        "type_id": "principal_standard",
        "address": "SP36ADRBVM8J00ZWR5QXC8V65WTJNCD1BF4EJ93ZZ"
      }
    }
  ],
  "anchor_mode": "any",
  "is_unanchored": false,
  "block_hash":
  "0x4f957ac52af57196eea8ae6ca9e848fc7772da72365d9ed4d3452afddc7a3cf2",
  "parent_block_hash":
  "0x4b7ea97418fd44fbc4af278424fbde67fbfb253398628f78319671d1eab48a47",
  "block_height": 38231,
  "burn_block_time": 1637173532,
  "burn_block_time_iso": "2021-11-17T18:25:32.000Z",
  "parent_burn_block_time": 1637173325,
  "parent_burn_block_time_iso": "2021-11-17T18:22:05.000Z",
  "canonical": true,
  "tx_index": 27,
  "tx_status": "success",
  "tx_result": {
    "hex": "0x0701000000000000000000000000000000000000000000000000",
    "repr": "(ok u9)"
  },
  "microblock_hash": "",
  "microblock_sequence": 2147483647,
  "microblock_canonical": true,
}
```

```

"events": [],
"execution_cost_read_count": 13,
"execution_cost_read_length": 4320,
"execution_cost_runtime": 5609000,
"execution_cost_write_count": 5,
"execution_cost_write_length": 21,
"tx_type": "contract_call",
"contract_call": {
  "contract_id":
"SPZW30K9VG6YCPYV4BX4V1FT0VJ66R1Q01W9DQ1W.nebula",
  "function_name": "claim",
  "function_signature": "(define-public (claim ))"
}
},

```

Subscribe via WebSockets:

```

client.subscribeTxUpdates('0xd78988664aaa9a1b751cd58c55b253914f79
0e95ca6f3d402a866559e1cbe0b3', event => {});

```

Subscribe via Socket.io:

```

sc.subscribeTransaction('0xd78988664aaa9a1b751cd58c55b253914f790e
95ca6f3d402a866559e1cbe0b3');

```

Address Transaction Updates

Sent every time a transaction is sent or received by a specific Stacks address. Transactions of different types and structures may be received.

Example message if subscribed to updates for an address

SP3C5SSYVKPAWTR8Y63CVYBR65GD3MG7K80526D1Q:

```

{
  "tx": {
    "tx_id":
"0x0c818b9af6356a2eb4d64ee1b2490193d97a82392c02e7264e006ae5979aa7
26",
    "nonce": 32,
    "fee_rate": "3000",
    "sender_address":
"SP3BK1NNSWN719Z6KDW05RBGVS940YCN6X84STYPR",
    "sponsored": false,
    "post_condition_mode": "deny",
    "post_conditions": [
      {

```



```

    "condition_code": "sent_equal_to",
    "amount": "4375722",
    "principal": {
      "type_id": "principal_contract",
      "contract_name": "newyorkcitycoin-core-v1",
      "address": "SP2H8PY27SEZ03MWRKS5XABZYQN17ETGQS3527SA5"
    }
  ],
  "anchor_mode": "any",
  "is_unanchored": false,
  "block_hash":
"0xe2a811451fed35331cf462a9107e3453fdebba1682dfad83cbbcdc603f644e
d3",
  "parent_block_hash":
"0x6d8653da23188d4d78ab9b6448229be68abe1bca001f8c574c094289107bce
15",
  "block_height": 58775,
  "burn_block_time": 1651720813,
  "burn_block_time_iso": "2022-05-05T03:20:13.000Z",
  "parent_burn_block_time": 1651720368,
  "parent_burn_block_time_iso": "2022-05-05T03:12:48.000Z",
  "canonical": true,
  "tx_index": 36,
  "tx_status": "success",
  "tx_result": {
    "hex": "0x0703",
    "repr": "(ok true)"
  },
  "microblock_hash": "",
  "microblock_sequence": 2147483647,
  "microblock_canonical": true,
  "event_count": 1,
  "events": [],
  "execution_cost_read_count": 15,
  "execution_cost_read_length": 31147,
  "execution_cost_runtime": 81975,
  "execution_cost_write_count": 2,
  "execution_cost_write_length": 123,
  "tx_type": "contract_call",
  "contract_call": {
    "contract_id":
"SP2H8PY27SEZ03MWRKS5XABZYQN17ETGQS3527SA5.newyorkcitycoin-core-
v1",
    "function_name": "claim-stacking-reward",
    "function_signature": "(define-public (claim-stacking-
reward (targetCycle uint)))",

```


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